

Elastic Dallas User Group

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Multimodal E-Commerce Search with Elastic and Jina

How e-commerce search went from BM25 to text embeddings to image embeddings, and what it takes to run it

Where e-commerce search started: BM25

Shopper types

“grey joggers for working out”

Catalog says

“Grey Melange Track Pants — Sports”

0 results

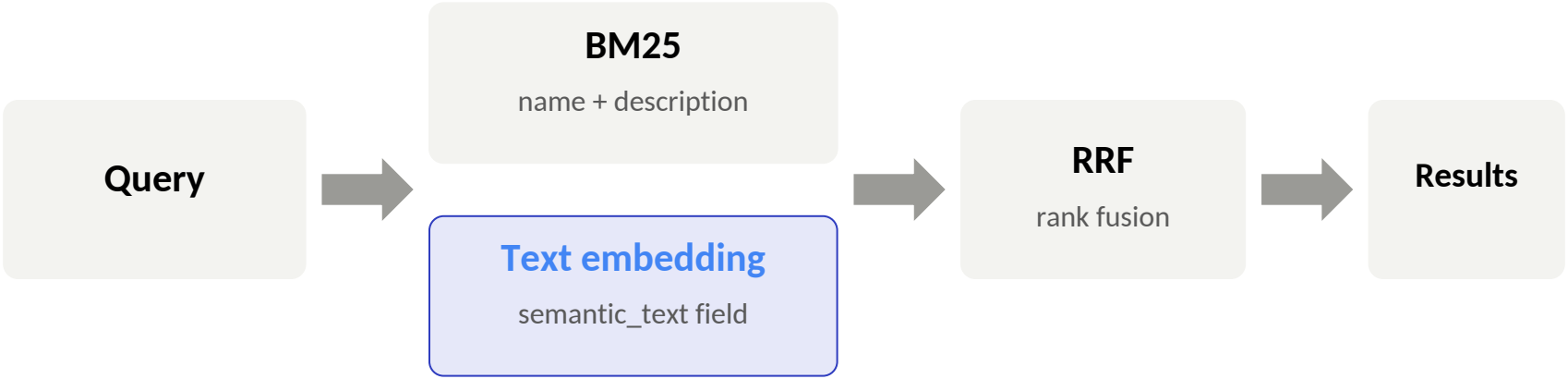
Still the backbone

- Exact matches on SKUs, brands, model names
- Fast, cheap, explainable scores
- TF/IDF matches on descriptions
- Filters, facets and boosts customers already rely on

Where it breaks

- Vocabulary gap: joggers vs track pants, tee vs t-shirt
- Descriptive intent: “gift for father’s day”
- Other languages than the catalog was written in

Step one: add text embeddings, keep BM25



What changes for the team

- Mapping, not pipeline: semantic_text embeds at ingest and at query time
- RRF fuses ranks, so BM25 and cosine/dot scores never have to be on the same scale
- Synonyms, paraphrase and cross-language queries start matching

Text embedding models in customer stacks

ELSER

Elastic sparse model. English, strong out of the box, no tuning. The default semantic_text choice for years.

EIS

jina-embeddings-v5-text-small

677M params, 119+ languages, 32K tokens, Matryoshka truncation. Highest MTEB among multilingual models under 1B.

EIS

E5 multilingual

Open model, runs on ML nodes. Common where data can't leave the cluster.

ML node

OpenAI text-embedding-3

Large/small variants via the inference API. Often already in the customer's AI stack.

External

Cohere Embed

Multilingual, int8/binary outputs, popular for RAG-first teams.

External

Google / Vertex embeddings

Chosen by GCP-centric customers; wired through the inference API.

External

What the description can’t see, the photo can

1.5×

more often ranks the right product first when image and text embeddings are combined, vs image alone

0.085 vs 0.073

Recall@1 on visual attributes (colour, pattern, sleeve) for image+text vs text only

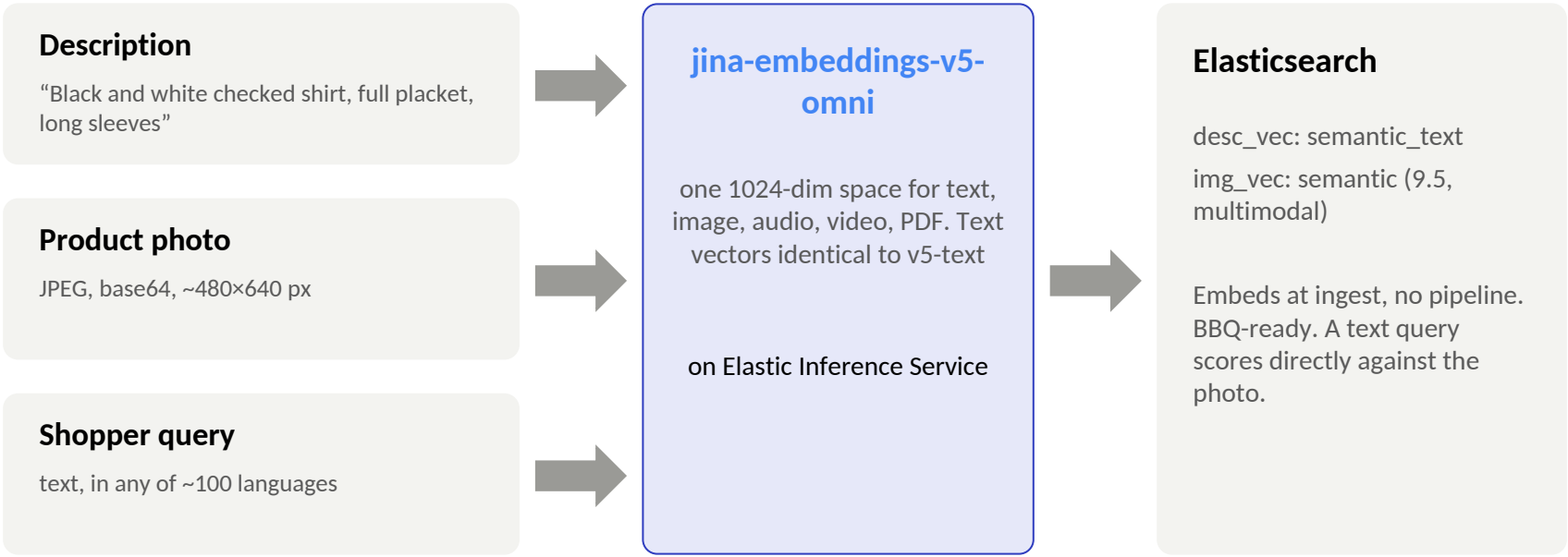
0.074 vs 0.065

Recall@1 for German queries over an English catalog, image+text vs text only

Where images help most: silhouette, colour, pattern, footwear shape. Where they add nothing: fabric, fit details only the copy mentions.

Elastic Search Labs benchmark, 5,000 apparel and footwear products, Jina embedding models, July 2026. The gap between strategies is the finding.

Step two: the photo joins the vector space



Third retriever in the same RRF: BM25 + text embedding + image embedding

semantic_text: Search simplified in Elasticsearch

1 Automatic Chunking Strategy

Eliminates complex manual chunking pipelines. Long documents are automatically chunked and processed seamlessly behind the scenes.

2 No Nested Docs Complexity

Simplifies mapping and querying significantly. Pass raw text directly without configuring nested fields or writing inner_hits queries.

3 Unified Text & Image Fields

Handles both text and multimodal image embeddings natively. Replaces separate explicit dense_vector type definitions.

4 Default Quantization

Applies scalar quantization out-of-the-box to drastically reduce memory usage and vector index footprint without manual tuning.

EIS: the models without the ML nodes

No model hosting

Embedding, rerank and LLM inference run on Elastic-managed GPUs. No ML nodes to size, patch or scale. Preconfigured endpoints appear in every deployment.

Throughput you don't have to engineer

Per-model requests-per-minute and tokens-per-minute limits are published; ingest throughput beats ML nodes, search latency is equivalent. Image inference runs on the same path.

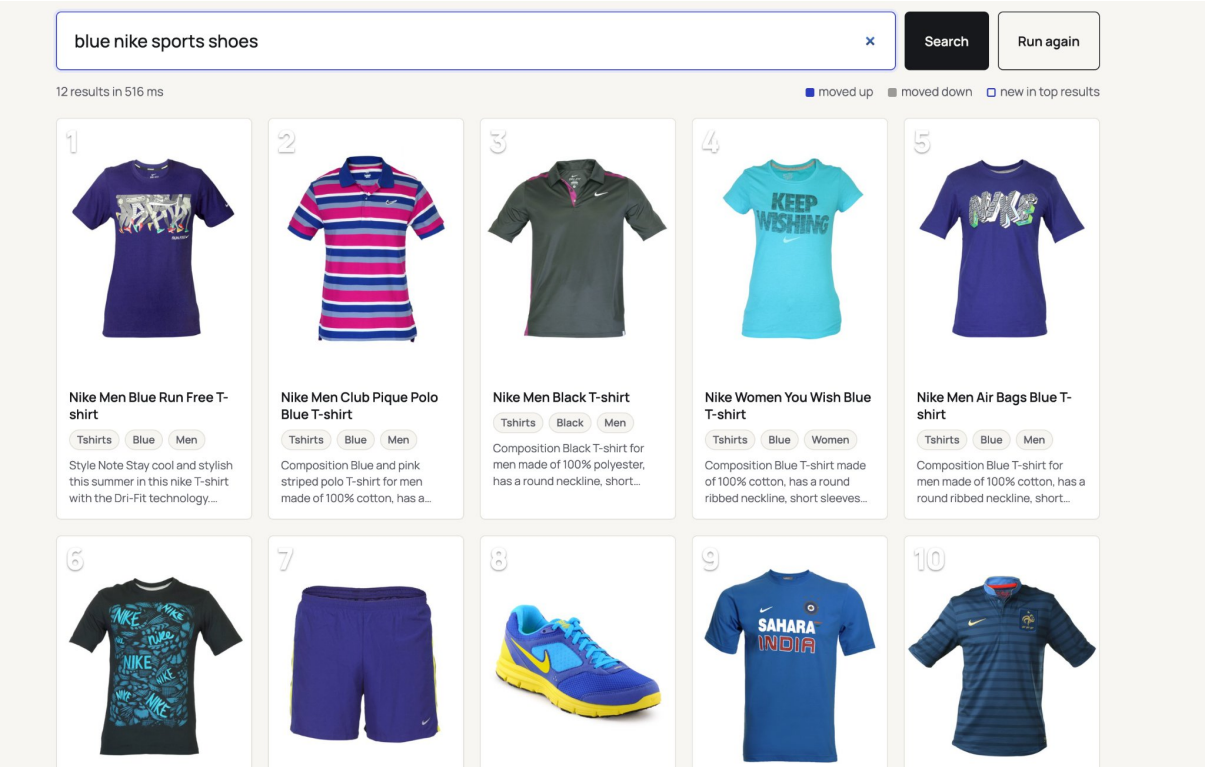
Pay per token

Billed per million tokens; embedding models bill input tokens only. No idle GPU, no capacity planning for the Black Friday reindex.

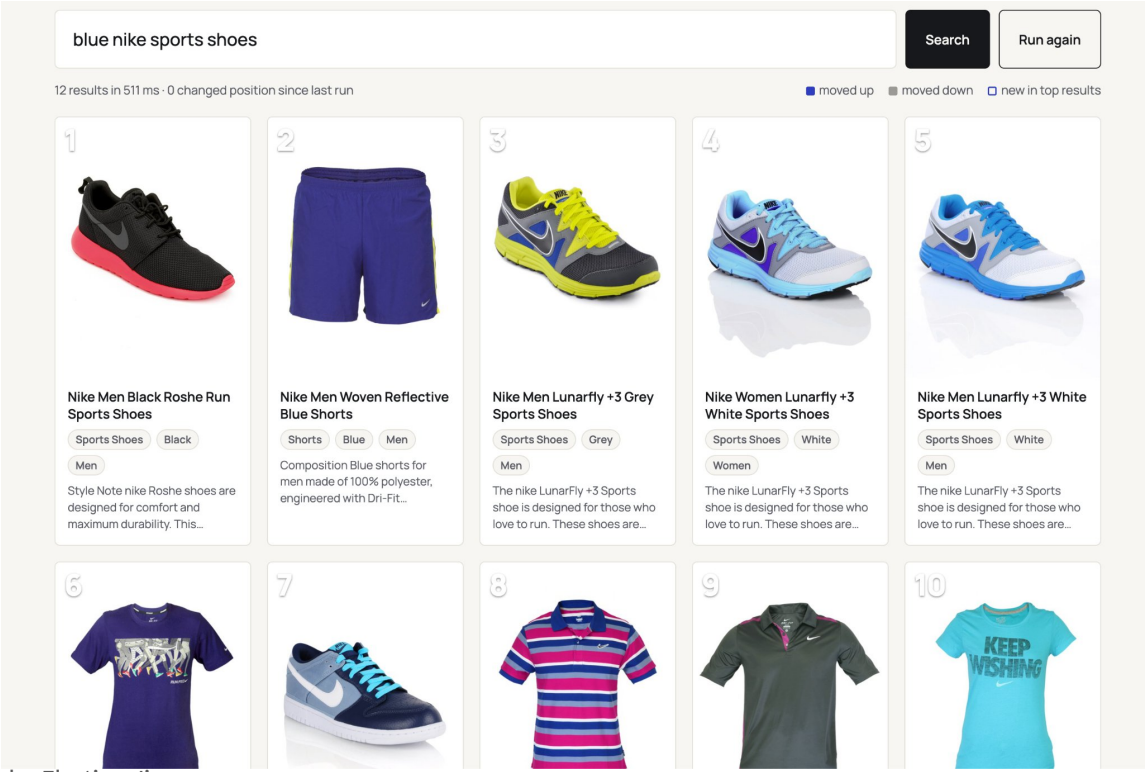
One catalog, every deployment

ELSER, Jina v5 text and omni, jina-clip-v2, Jina rerankers, plus OpenAI, Anthropic and Google LLMs. ECH, Serverless, and self-managed through Cloud Connect.

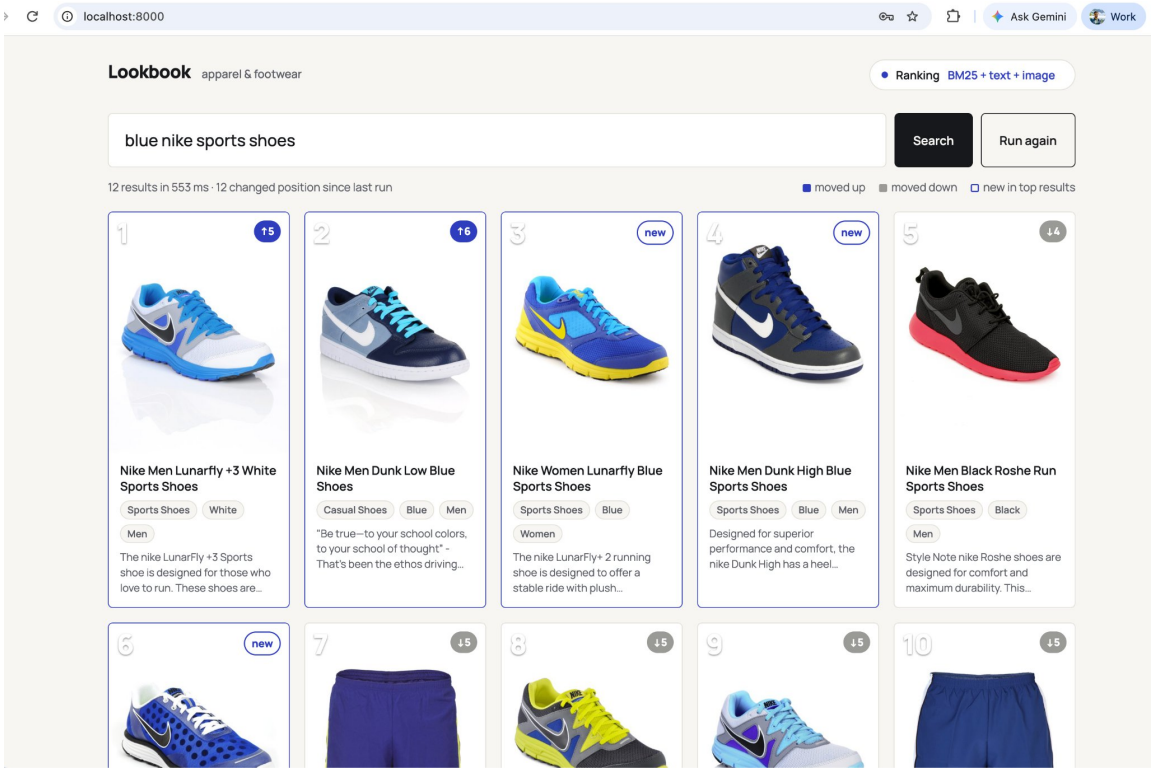
Demo: BM25 match query on name and description



Demo: BM25 + description text embeddings RRF



Demo: Bm25 + text + image RRF



What to take back to your catalog

1 Keep BM25 & Embeddings

Keep BM25. Add text embeddings first, image embeddings where the photo carries the signal.

2 Filters vs Ranking

Hard constraints stay filters: gender, category, size, stock. Embeddings rank, they never gate.

3 Benchmark Locally

Benchmark on your own products and queries

4 Fill Gaps with AI

Use generated descriptions to fill gaps, not to replace copy written by people.

How to get involved

1 Present a Talk

Interested in presenting? We would love to hear from you. We welcome lightning talks, deep dives, and everything in between.

2 Grow Our Community

Help us grow our user groups! Bring a friend or colleague to our next meetup event.

3 Host an Event

Is your company interested in hosting? Hosting a meetup is a great recruiting opportunity, and Elastic covers pizza, beer, and swag!

4 Partner With Us

Know an affinity meetup chapter we could partner with for a joint event? Let us know—we love collaborating with local tech groups.

Something else? Feel free to reach out to us to chat more:
meetups@elastic.co

